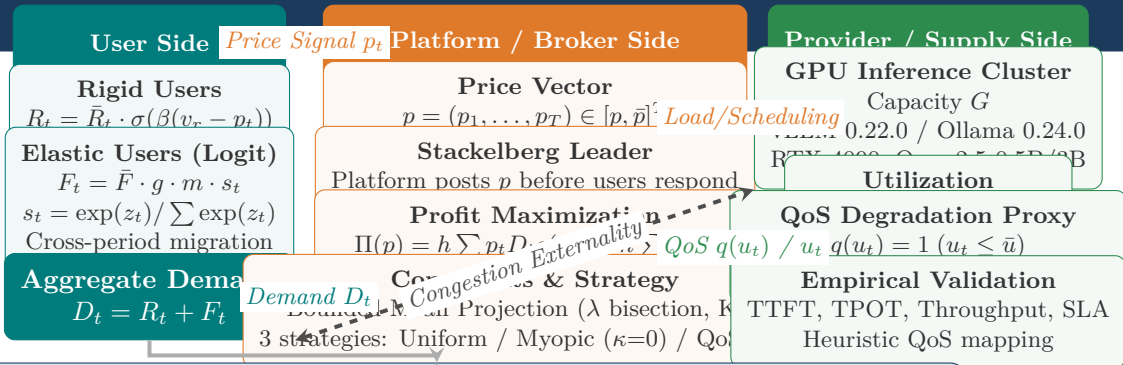


Token Service Cross-Period Pricing & Demand Response Game Framework



Algorithm & Game Layer

Game Structure

- Stackelberg Leader-Follower: platform posts
- Congestion externality: GPU overload $\rightarrow$ QoS
- Welfare decomposition: profit + rigid surplus

Numerical Optimization

- Multi-start SLSQP + equality/bound/bill-protection constraints
- Bounded Mean Projection: λ bisection, KKT optimal
- Candidates sorted by profit; residuals $< 10^{-12}$
- Scalability: $T = 8 \rightarrow 64$, $5 \rightarrow 40$ starts

Key Results: QoS-aware vs Myopic +12.53% · QoS floor 0.6632 $\rightarrow$ 0.9918